

Project 4: Unsupervised Image Clustering and Visualization

(K-Means Clustering, PCA and t-SNE on the MNIST Handwritten Digits)

1. Executive Summary

This project uses unsupervised learning to find natural groups in 70,000 handwritten digit images (MNIST) without using their labels. K-Means clustering grouped the images into 10 clusters, which were then evaluated against the true digits, tested on new images, and visualized in 2D with PCA and t-SNE.

Question	Answer
Best number of clusters?	No sharp elbow; k = 10 chosen (gains flatten near k = 8-10; matches the 10 digits)
How well do clusters match digits?	~ 59% accuracy (majority mapping); ARI 0.367 , NMI 0.498 on the test set
Does the model generalize?	Yes. Mean distance 6.161 (train) vs 6.163 (test); 95.2% of test images fit their cluster
Hardest digits?	5 and 9 never get their own cluster (0% accuracy); 4/7/9 and 3/5/8 get mixed
Easiest digits?	1 (99.5%), 0 (88.9%), 6 (83.0%)
Best 2D view?	t-SNE (trustworthiness 0.982 vs 0.739 for PCA)
Biggest improvement found?	K-Means on the t-SNE map: accuracy 58.0% → 81.9% , ARI 0.364 → 0.713

Main lesson: the digits are clearly separable, but not by straight-line pixel distance. How the data is represented matters more than the clustering algorithm itself.

2. Objective and Data

2.1 Objective

Discover groups in the MNIST images with K-Means, judge how meaningful those groups are, show how the trained model assigns new images, and use dimensionality reduction to visualize and assess the clustering. Labels were used **only for evaluation**, never for training.

2.2 Datasets

	MNIST (main)	load_digits (external test)
Images	70,000 (60,000 train / 10,000 test)	1,797
Resolution	28 x 28	8 x 8 (upscaled to 28 x 28)
Features	784	64
Raw pixel range	0-255	0-16

	MNIST (main)	load_digits (external test)
Source	fetch_openml("mnist_784")	sklearn.datasets.load_digits

Tools: Python 3, NumPy, pandas, scikit-learn (KMeans, PCA, TSNE, metrics), SciPy, matplotlib, seaborn, Plotly, Pillow and joblib.

3. Methodology

3.1 Data Preparation

- Loaded the data with `fetch_openml` and kept the standard 60,000 / 10,000 train-test split.
- Checked quality with assert statements: no missing values, shape 70,000 x 784, pixels within 0-255.
- Normalized pixels to 0-1 (divide by 255) so all features share one scale. This matters because K-Means uses Euclidean distance.
- Flattened each 28 x 28 image into a 784-value vector and verified the round trip.

3.2 Exploratory Data Analysis

Class balance, sample images, mean image per digit, pixel intensity distribution, pixel variance map, ink per digit, and PCA explained variance.

3.3 Clustering

- Searched $k = 2$ to 15 on a 10,000-image sample using the elbow method (inertia) and the silhouette score (on a 3,000-image subset).
- Trained the final K-Means model ($k = 10$, $n_init = 10$, $random_state = 42$) on all 60,000 training images.
- Interpreted clusters with centroid images, a cluster-vs-digit table, purity, and sample images per cluster.

3.4 Evaluation and Prediction

- Built reusable functions: `preprocess_images()` for any input shape, `predict_cluster()` returning cluster, distance, runner-up and confidence margin, and `to_mnist_format()` for real image files.
- Defined a per-cluster "good fit" threshold at the 95th percentile of training distances.
- Validated on the unseen test set, the external `load_digits` set, noisy images ($\sigma = 0.1$ to 0.5) and a custom image file.
- Scored with majority-mapping accuracy, Adjusted Rand Index (ARI), Normalized Mutual Information (NMI) and a confusion matrix.

3.5 Dimensionality Reduction and Visualization

- PCA to 2D for plotting and to 50D as input for t-SNE; t-SNE to 2D (perplexity 30) on a 5,000-image sample.
- Compared views with silhouette score and trustworthiness; mapped mismatches and low-confidence points; analyzed centroid distances and a dendrogram.

- Tested t-SNE perplexity 5, 30 and 50, and re-ran K-Means inside each reduced space.

4. Key Decisions

Decision	Reason
Normalize pixels to 0-1	Puts every feature on one scale for distance-based K-Means
Keep the official 60k / 10k split	Standard benchmark; test images stay truly unseen
Sample 10k (elbow), 3k (silhouette), 5k (t-SNE)	Large time savings while staying representative
k = 10	No sharp elbow; gains flatten near 10; matches the 10 digit classes
Majority mapping for scoring	Simple, explainable way to compare clusters with labels
Per-cluster 95th percentile fit threshold	Clusters differ in spread (7.07 to 8.35), so one global cut-off would be unfair
PCA to 50D before t-SNE	Standard practice; removes noise and speeds up t-SNE
random_state = 42 throughout	Reproducible results
Save model, mapping and thresholds	Predictions work later without retraining

5. Analysis and Results

5.1 Exploratory Data Analysis

- **Balanced classes:** each digit is 8.9% to 11.4% of the data, so no resampling was needed.
- **Sparse images:** about 81% of pixels are background, and 67 of 784 pixels are zero in every training image.
- **Redundant features:** PCA keeps 80% of the variance with 44 components, 95% with 154 and 99% with 331.
- **Ink varies by digit:** 1 uses the least ink (mean 59.6); 0 uses the most (135.9).

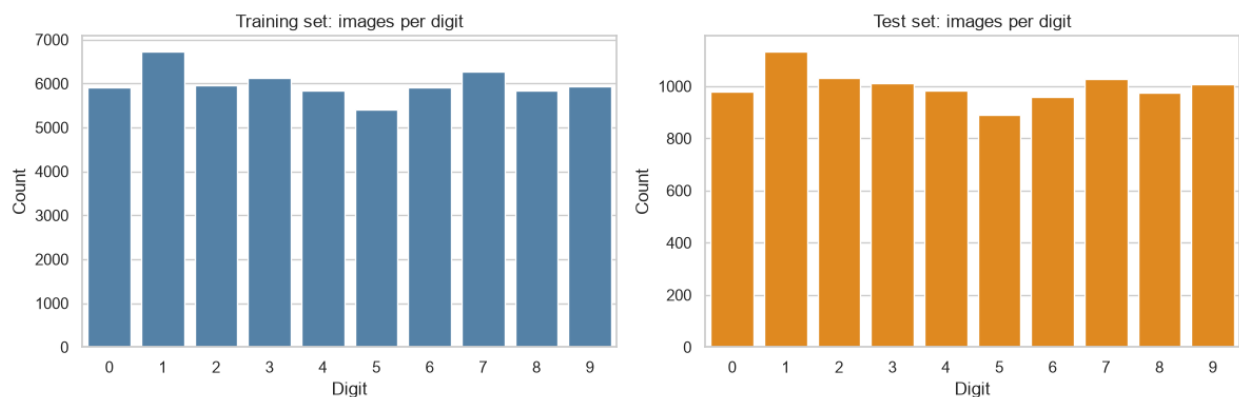


Figure 1. Images per digit in the training and test sets.

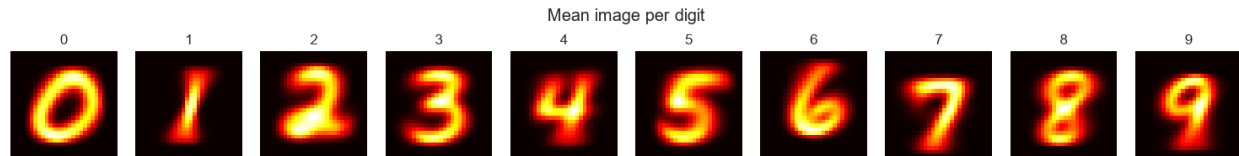


Figure 2. Mean image per digit.

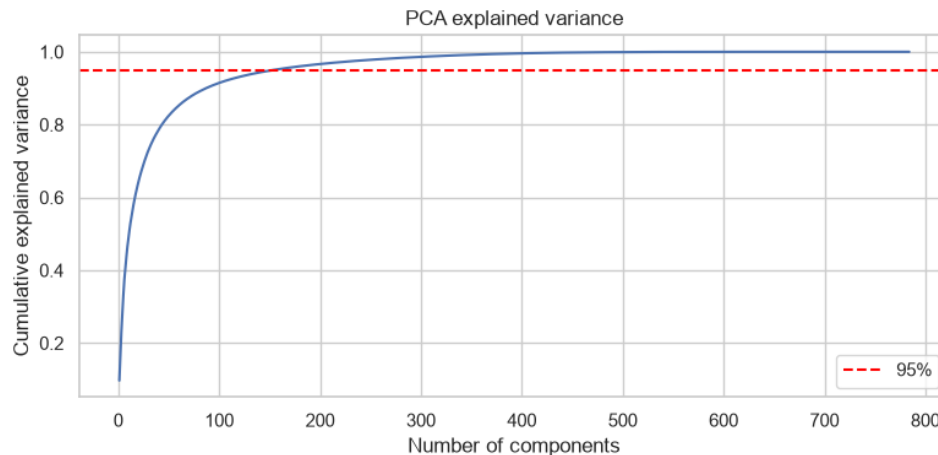
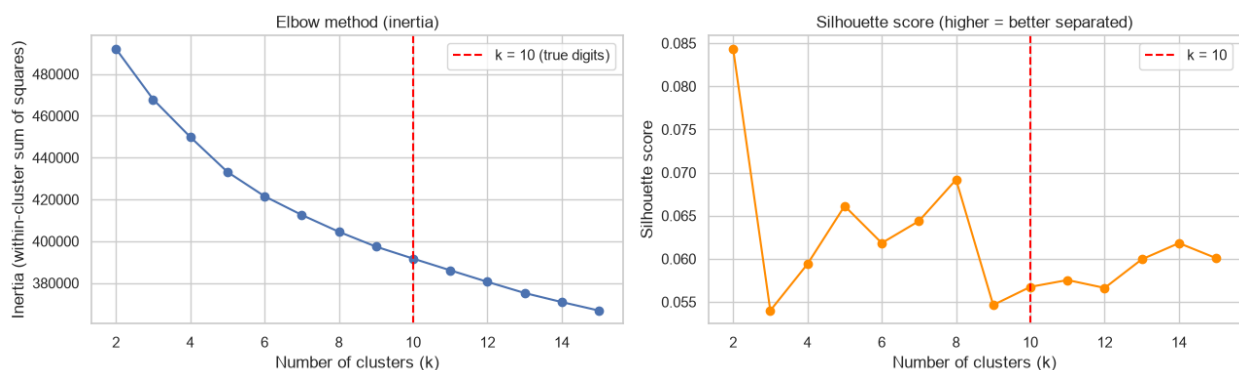


Figure 3. Cumulative PCA explained variance.

5.2 Choosing the Number of Clusters

The elbow curve bends gradually with **no sharp elbow**. The inertia drop per extra cluster falls from 4.9% at $k = 3$ to about 1.4% from $k = 10$ onward. Silhouette scores are very low for every k (0.05 to 0.08), which is typical for raw pixel data: in 784 dimensions all images are fairly far apart and digits blend into each other. The highest silhouette is at $k = 2$, which is too coarse to be useful. **$k = 10$** was chosen because the curve flattens there and it matches the known number of digits.

Figure 4. Elbow method (left) and silhouette score (right) for $k = 2$ to 15.

5.3 Cluster Interpretation

The final model trained in 45.5 seconds (inertia 2,352,828). Cluster sizes range from 3,098 to 8,929 images even though the digits are balanced, which is the first sign that clusters do not map one-to-one to digits.

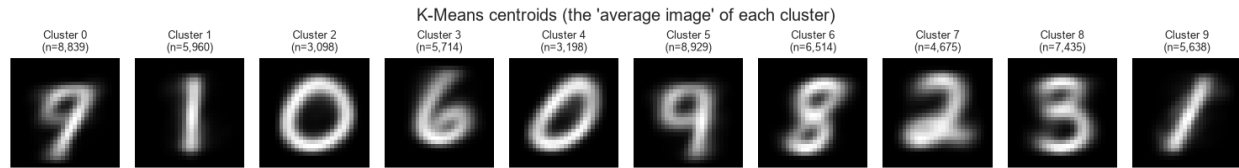


Figure 5. K-Means centroids: the average image of each cluster.

Cluster	Size	Majority digit	Purity	Second digit	Avg ink
C2	3,098	0	90.7%	6 (3.2%)	139.4
C4	3,198	0	78.1%	5 (8.6%)	139.5
C1	5,960	1	62.4%	3 (7.1%)	61.0
C9	5,638	1	52.8%	5 (14.3%)	71.6
C7	4,675	2	89.7%	3 (4.6%)	124.3
C8	7,435	3	52.8%	5 (23.8%)	114.7
C5	8,929	4	35.7%	9 (32.5%)	95.1
C3	5,714	6	86.0%	2 (3.7%)	109.9
C0	8,839	7	42.7%	9 (27.8%)	94.1
C6	6,514	8	53.1%	5 (22.7%)	116.5

- **Digits split by writing style:** C2 is a round, upright 0 and C4 a slanted 0; C1 is an upright 1 and C9 a slanted 1.
- **Look-alike digits share clusters:** C5 is almost equally 4 and 9; C0 mixes 7 and 9; C8 and C6 each absorb many 5s.
- **5 and 9 have no cluster of their own.** Because 0 and 1 used two clusters each, only eight clusters were left for the other eight digits.
- **Ink drives part of the grouping:** the 1-clusters have the lowest average ink, the 0-clusters the highest.

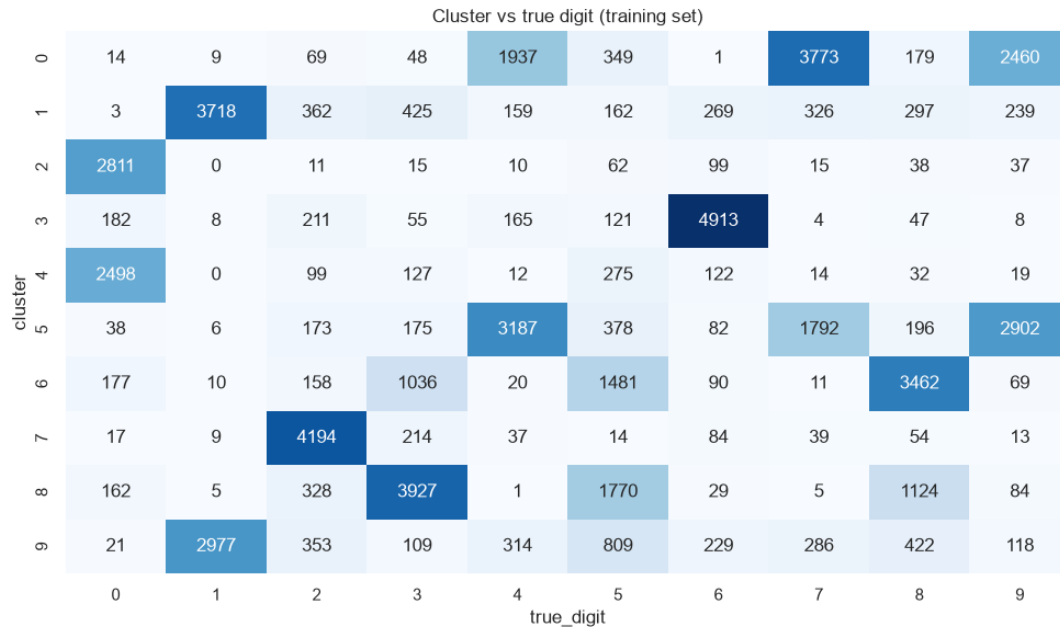


Figure 6. Cluster vs true digit (training set).

5.4 Evaluation and Validation

Each cluster was labeled with its majority digit to score the model like a classifier. Train accuracy was **59.1%** and test accuracy **59.5%**, with test ARI **0.367** and NMI **0.498**: far above random (10%), but well below a supervised classifier.

Dataset	Images	Accuracy	Good fit	Mean distance	Mean margin
Train (reference)	60,000	59.1%	95.0%	6.161	0.145
Test (unseen MNIST)	10,000	59.4%	95.2%	6.163	0.144
load_digits (external)	1,797	33.6%	100.0%	5.028	0.121

Generalization: test images fit the clusters just like training images. Per-cluster mean distances change by only $\pm 1.6\%$, and cluster shares match closely. The $\sim 59\%$ accuracy is a limit of K-Means on pixels, not overfitting.

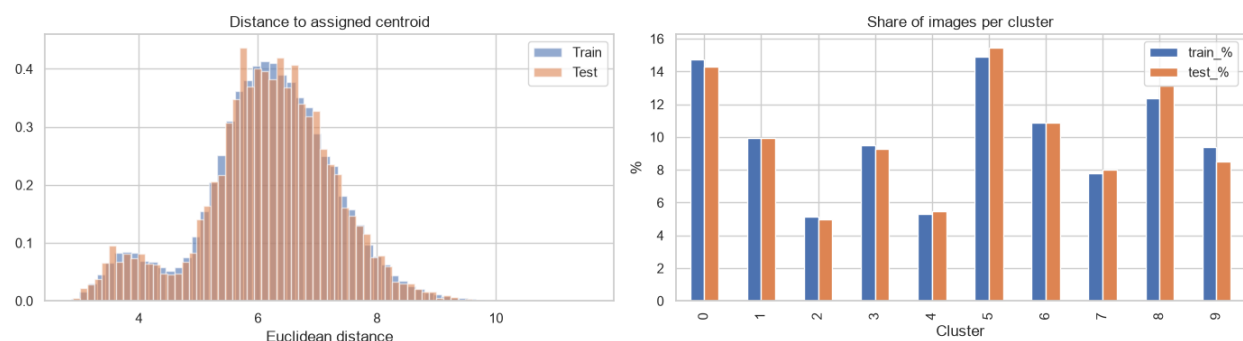


Figure 7. Distance to assigned centroid (left) and share of images per cluster (right), train vs test.

Per-digit accuracy (test): 1 (99.5%), 0 (88.9%), 6 (83.0%), 3 (68.9%), 2 (68.5%), 8 (60.2%), 7 (58.6%), 4 (56.9%), 5 (0%), 9 (0%). Digits 5 and 9 are structurally impossible to predict because no cluster has them as its majority.

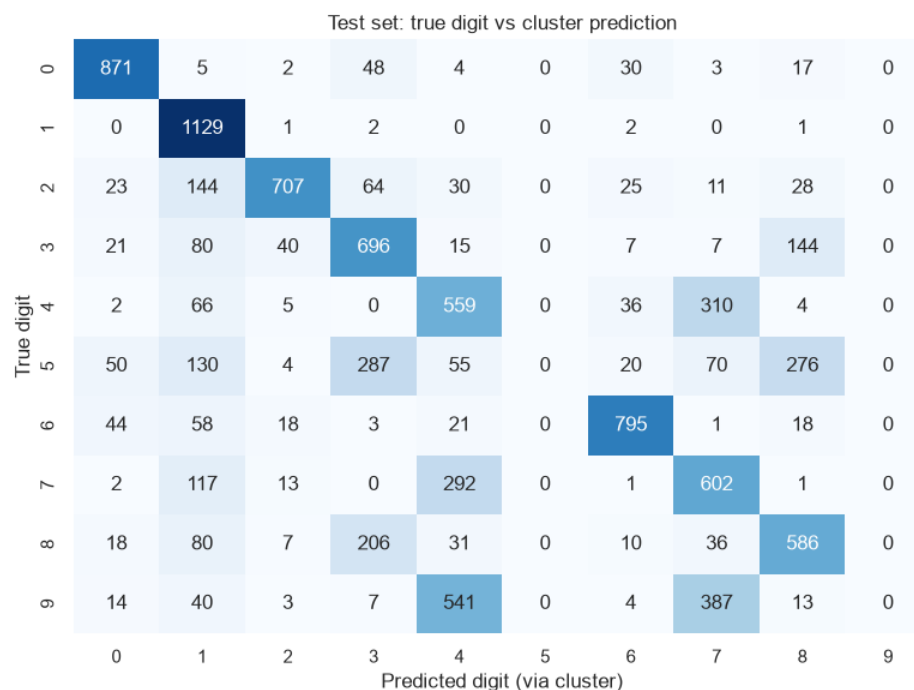


Figure 8. Test set: true digit vs cluster prediction.

External data: accuracy on load_digits drops to 33.6%, yet 100% of images count as a good fit. Upscaled 8 x 8 images are smooth and blurry, so they sit close to the (also blurry) centroids even when the cluster is wrong. The good-fit flag measures distance, not correctness, and cannot detect this kind of style shift.

Noise robustness:

Noise σ	Same cluster as clean	Accuracy	Good fit
0.0	100.0%	59.4%	95.2%
0.1	97.2%	60.0%	96.9%
0.2	94.2%	60.8%	93.8%
0.3	91.1%	61.1%	67.4%
0.5	83.9%	60.2%	0.0%

Cluster assignments stay stable under noise and accuracy stays near 60%. Unlike the external case, the fit check does catch heavily noisy images (0% good fit at $\sigma = 0.5$).

Custom image: a 7 saved as dark ink on white paper at 200 x 200 px was converted with `to_mnist_format()` and correctly assigned to cluster 0 (digit 7) with a margin of 0.33, well above the average of about 0.14.

5.5 Dimensionality Reduction and Visualization

Two PCA components keep only **16.8%** of the variance (PC1 9.7%, PC2 7.1%), while 50 components keep 82.5%. In the PCA view the digits overlap in one cloud; in the t-SNE view they form clear, separate islands.

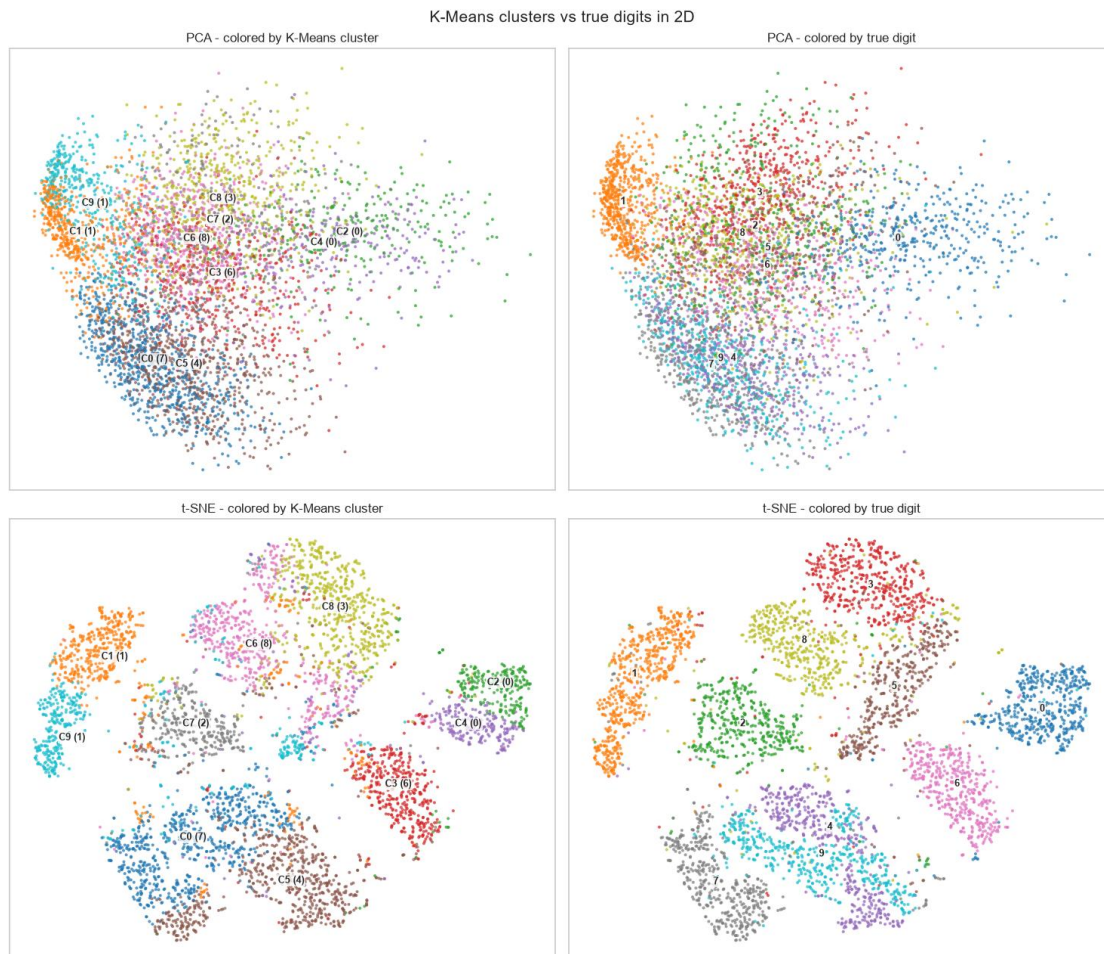


Figure 9. PCA (top) and t-SNE (bottom), colored by K-Means cluster (left) and true digit (right).

View	Silhouette (clusters)	Silhouette (true digits)	Trustworthiness
Original (784D)	0.059	0.046	1.000
PCA (2D)	-0.019	0.036	0.739
t-SNE (2D)	0.185	0.316	0.982

- t-SNE keeps 98% of real neighborhoods; PCA places many unrelated images next to each other.
- In t-SNE space the **true digits separate better (0.316)** than the **K-Means clusters (0.185)**: the structure is in the data, but K-Means draws the boundaries in the wrong places.

Where K-Means goes wrong: 41% of the 5,000 sampled points do not match their cluster's majority digit. The largest mix-ups are $9 \rightarrow 4$ (248), $9 \rightarrow 7$ (226), $4 \rightarrow 7$ (187), $7 \rightarrow 4$ (148), $5 \rightarrow 3$ (141), $5 \rightarrow 8$ (124) and $3 \rightarrow 8$ (92), forming two families of similar shapes: **{4, 7, 9}** and **{3, 5, 8}**.

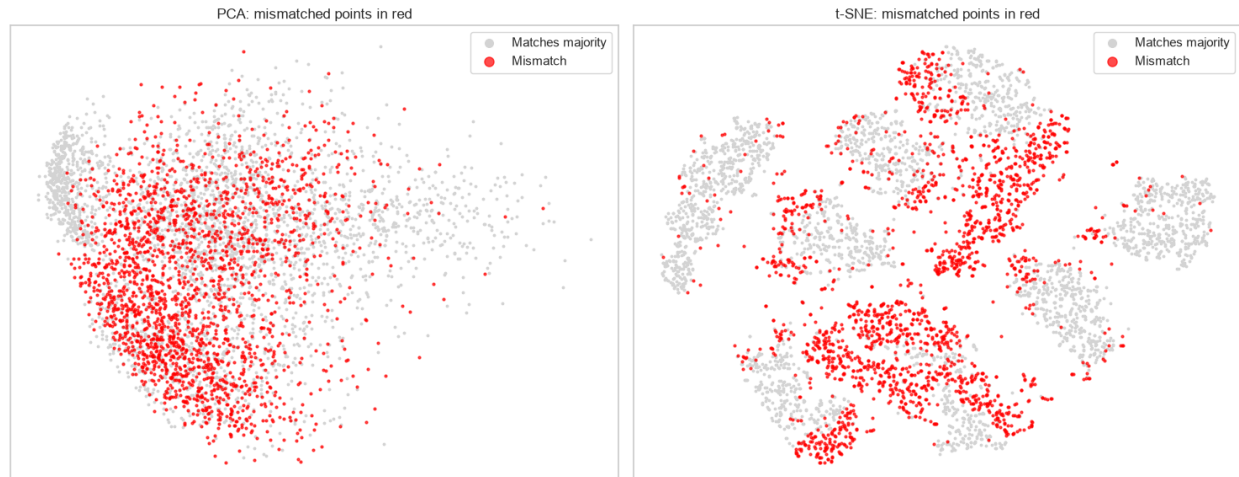


Figure 10. Points whose cluster majority does not match the true digit (red).

Per-cluster separation: C7 (2), C3 (6) and C2 (0) are the best separated. C4 (0) and C9 (1) have negative scores because upright and slanted versions of the same digit share one t-SNE island.



Figure 11. Mean t-SNE silhouette per cluster.

Relationships between clusters: the closest centroid pairs are C0 (7) ↔ C5 (4) at 3.98, C1 (1) ↔ C9 (1) at 4.02, C8 (3) ↔ C6 (8) at 4.24 and C7 (2) ↔ C3 (6) at 4.46. Among the least confident 10% of points, the most common competing pairs are C0 ↔ C5 (64 points) and C6 ↔ C8 (55 points), and these points sit on the borders between t-SNE islands.

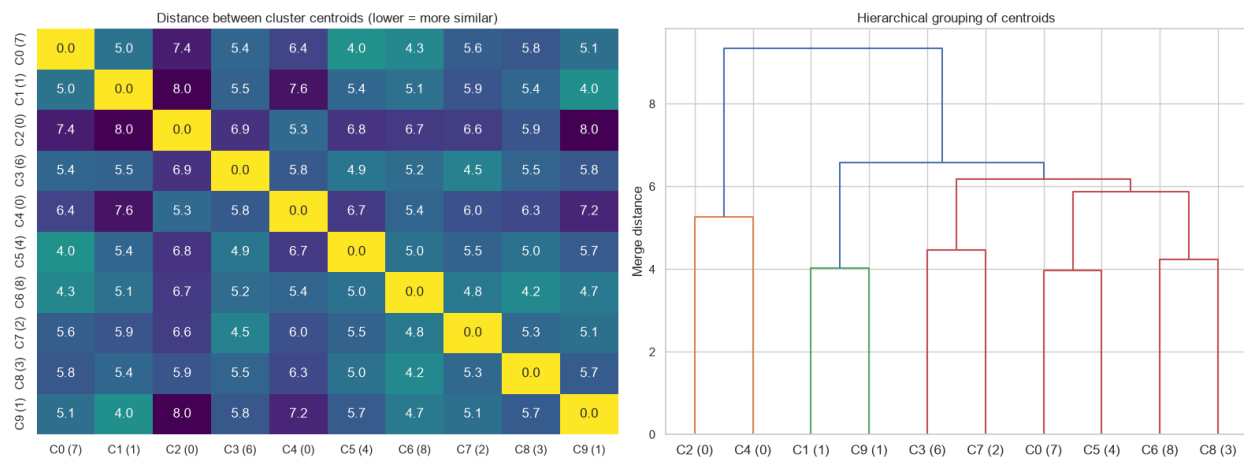


Figure 12. Distance between centroids (left) and hierarchical grouping of centroids (right).

Perplexity check: perplexity 5 breaks the islands into fragments, while 30 and 50 give clean, similar layouts, so the conclusions do not depend on one setting.

Clustering in reduced space:

Space	Accuracy	ARI	NMI
Pixels (784D)	58.0%	0.364	0.494
PCA (50D)	55.7%	0.341	0.478
PCA (2D)	41.5%	0.247	0.367
t-SNE (2D)	81.9%	0.713	0.767

PCA does not help, because it keeps the same linear structure K-Means already sees. t-SNE unfolds the curved digit shapes into compact islands, which nearly doubles ARI. Because t-SNE cannot transform new images, this is an analysis result rather than a deployable model.

6. Insights

1. **K-Means finds real but partial structure.** About 59% accuracy without labels is far above random, but it does not replace supervised learning.
2. **It groups by appearance, not meaning.** It spends two clusters on slanted vs upright 0s and 1s while look-alike digits share a cluster.
3. **Two families cause most errors: {4, 7, 9} and {3, 5, 8}.** The confusion matrix, centroid distances, dendrogram and boundary analysis all agree.
4. **The model generalizes within MNIST.** Train and test behave almost identically.
5. **"Good fit" is not "correct".** The external data was 100% good fit but only 33.6% accurate. Distance checks catch noise but miss style shifts.
6. **t-SNE is the better view.** It keeps 98% of local neighborhoods; 2D PCA keeps only 17% of the variance and can mislead.

7. **The distance measure is the bottleneck.** Clustering in t-SNE space reaches 81.9% accuracy, so the digits are separable, just not by pixel distance.
8. **The confidence margin is useful.** Low-margin images sit on cluster borders and are the best candidates for human review.

7. Limitations

- k was chosen partly with domain knowledge; the metrics alone did not point clearly to 10.
- Majority mapping needs labels and hides digits that are never a majority (5 and 9).
- Samples were used for the k search, silhouette and t-SNE, so results could shift slightly with another sample.
- t-SNE cannot transform new images, and distances between its islands are not meaningful.
- K-Means assumes round, similar-sized clusters, which handwritten digits do not have.
- The fit threshold is distance-based and does not detect style or domain shift.

8. Recommendations and Next Steps

- **Use more clusters (k = 15-20)** and map several clusters to each digit so 5 and 9 can appear.
- **Cluster in a non-linear embedding that handles new data**, such as UMAP or an autoencoder.
- **Compare other algorithms:** Gaussian Mixture Models, DBSCAN/HDBSCAN and Agglomerative clustering.
- **Use the margin for human-in-the-loop labeling**, sending the least confident images for review.
- **Monitor new data sources with a small labeled sample**, since distance alone missed the external-data drop.

9. Conclusion

K-Means on normalized MNIST pixels found ten meaningful groups that match the real digits about 59% of the time, behave the same on unseen images, and stay stable under noise. The analysis shows why it falls short: the method groups by pixel similarity, so it splits digits by writing style and merges look-alike digits (4/7/9 and 3/5/8). Dimensionality reduction confirmed this: t-SNE revealed clear digit structure (trustworthiness 0.982), and clustering in that space raised accuracy to 81.9%. The key lesson is that **how the data is represented matters more than the clustering algorithm itself.**

Appendix: Saved Outputs and Reproducibility

- **results/**: 21 CSV tables covering EDA, the elbow search, cluster profiles, fit thresholds, validation, the confusion matrix, per-image predictions, centroids and 2D coordinates for Power BI or Tableau.
- **models/kmeans_k10.joblib** (0.27 MB) and **models/model_info.json** (cluster-to-digit mapping, fit thresholds, preprocessing). A reload test confirmed identical predictions.
- All files are under GitHub's 100 MB limit.
- To reproduce: open the notebook and run **Kernel → Restart & Run All**. All random steps use seed 42.